MAT1011 Midterm Exam

MathPi

October 2023

Question 1 (15 marks)

Let

$$f(x) = (x-1)x^{5/3}.$$

(a) Find the derivatives $f'(x)$ and $f''(x)$ when they exist.

(b) Find the tangent line and normal line of the curve $y = f(x)$ at the point $P_0(0,0)$.

(c) Find all critical points of f.

Question 2 (25 marks)

Find the following limits.

(a)

$$\lim_{x\to 0} \frac{\sin^3\big(\sin^2 x\big)}{\sin^2\big(\sin(x^3)\big)}.$$

(b)

$$\lim_{x\to 0^+} \frac{\cos(1+\sqrt{x}) - \cos 1}{\sqrt{\sin x}}.$$

(c)

$$\lim_{x\to 0} \frac{1}{x^2} \int_0^{x^2} \left[\cos(t^2) + \cos^2(t^2)\right] dt.$$

Question 3 (10 marks)

Find the limit

$$\lim_{x\to 0} \frac{\sqrt{1+2x}-1}{x}.$$

Then use the ε-δ language to prove your result.

Question 4 (25 marks)

(a) Evaluate the indefinite integral

$$\int |x| \cos x \, dx.$$

(b) For any non-negative integer n denote

$$J_n = \int \sin^n x \, dx.$$

Show that for any $n \geq 2$ it holds that

$$J_n = \frac{n-1}{n} J_{n-2} - \frac{1}{n} \sin^{n-1} x \cos x.$$

(c) Evaluate the following:

$$\int \left[\frac{d}{dx} f(\cos^2 x) \right] dx - \frac{d}{dx} \int f(\cos^2 x) \, dx,$$

where f is a continuously differentiable function on $\mathbb{R}$.

Question 5 (18 marks)

(a) Let

$$u(x) = g(x) D(x),$$

where g is differentiable at $x = 0$ and $g(0) = g'(0) = 0$, and $D(x)$ is the Dirichlet function,

$$D(x) = \begin{cases} 0 & \text{if } x \text{ is irrational,} \\ 1 & \text{if } x \text{ is rational.} \end{cases}$$

Is $u(x)$ differentiable at $x = 0$? If your answer is "YES" then find $u'(0)$; if your answer is "NO" then give your reason.

(b) Let $f(x) = \cos(\zeta(x))$, where $\zeta(x)$ is the Riemann function,

$$\zeta(x) = \begin{cases} 0, & \text{if } x \text{ is irrational,} \\ \dfrac{1}{q}, & \text{if } x = \dfrac{p}{q}, \text{ an irreducible fraction.} \end{cases}$$

Is $f(x)$ integrable over the interval $[0, 1]$? If your answer is "YES" then find $\int_0^1 f(x) \, dx$; if your answer is "NO" then give your reason.

Question 6 (7 marks)

Assume $f(x)$ is a continuously differentiable function on the interval $[0, 1]$ and $f'(x) \neq 0$ for all $x \in [0, 1]$.

(a) Prove that there exists a unique number $\xi_* \in [0, 1]$ such that

$$\int_0^1 f(x) \, dx = f(\xi_*).$$

(b) For any positive integer n, let

$$S_n = \sum_{k=1}^{n} f\left(\frac{k}{n}\right) \frac{1}{n}.$$

Prove that there exists a unique $\xi_n \in [0,1]$ such that

$$S_n = f(\xi_n).$$

(c) Prove that for any $\varepsilon > 0$ there exists a positive integer $N = N(\varepsilon)$ such that the numbers ξ_* and ξ_n given in (a) and (b) above respectively satisfy

$$|\xi_n - \xi_*| < \varepsilon \quad \text{whenever } n > N.$$